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Method for laser welding two coated workpieces

Abstract

A method for laser welding two coated workpieces includes positioning an upper workpiece and a lower workpiece on top of each other and passing a first laser beam over the upper and lower workpieces from a side of the upper workpiece so as to at least partially evaporate the respective coating of each of the workpieces on their facing sides along a depletion trace. A second laser beam is passed over the workpieces from the side of the upper workpiece so as to melt a material of the two workpieces within the depletion trace, and thereby weld the workpieces to one another. In the first laser passing, the first laser beam melts the material of the upper workpiece, so that a web of non-melted material of the upper workpiece remaining between the melted material of the upper workpiece and the facing side of the upper workpiece.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of International Application No. PCT/EP2021/053305 (WO 2021/175555 A1), filed on Feb. 11, 2021, and claims benefit to German Patent Application No. DE 10 2020 105 505.6 filed on Mar. 2, 2020. The aforementioned applications are hereby incorporated by reference herein.

FIELD

(1) The invention relates to a method for laser welding two coated workpieces, in particular steel sheets coated with zinc, an upper workpiece and a lower workpiece being positioned one on top of the other, with, in a first method step, a laser beam being passed over the workpieces positioned one on top of the other from the side of the upper workpiece, and the coating of the workpieces on their facing sides at least partially evaporating along a depletion trace, and with, in a second method step, a laser beam again being passed over the workpieces positioned one on top of the other from the side of the upper workpiece, and the material of the two workpieces being melted within the depletion trace, as a result of which the workpieces are welded to one another.

BACKGROUND

(2) Such a method has been disclosed by DE 103 09 157 A1, for example.

(3) Welding is a joining method which enables two workpieces to be permanently connected to one another. Laser welding is usually used if the intention is to carry out welding at a high welding speed, with a narrow and slender weld seam shape and with low thermal warpage. During laser welding, energy is fed in via a laser beam.

(4) In many cases, the workpieces to be welded to one another are coated, for example in order to protect the workpieces against corrosion. In the automotive field, bodywork parts of sheet steel as base material are often coated with zinc, for example.

(5) In the event of laser welding coated workpieces, however, defects often arise (for instance pores and spatter) at the weld seam if the material of the coating has such a low evaporation point that it evaporates at the temperature required for welding the base material of the workpieces. Primarily critical is coating material which becomes involved in the welding process at the joint of workpieces positioned overlapping on one another (lying one on top of the other). Suddenly evaporating coating material in the melt pool creates welding spatter and can result in holes or porosity in the weld seam (what is referred to as the “melt pool effect”). In addition, a coating material gas flow can flow at high pressure from the joint into a vapor capillary of the welding process and press against the back wall of the vapor capillary, and thus distend the vapor capillary of the welding process (what is known as the “balloon effect”); this can likewise generate spatter and tunnel-like pores.

(6) A measure for reducing defects caused by coatings on workpieces is, during the laser welding, not to position the workpieces one on top of the other but rather to arrange them spaced apart one above the other. However, this requires complex preparation when the workpiece is being positioned, and can create surface defects in the form of indentations.

(7) To reduce defects in the weld seam when laser welding two plates positioned one on the other as far as possible without a gap, DE 103 09 157 A1 proposes using a laser beam first of all in a first method step to heat the coated plate facing the laser beam, with the result that the coating of the plates on their facing sides evaporates, with neither of the plates melting. In a second method step, the two plates can be welded in the region in which the coating has been removed. In this context, the same laser beam is used in the first and the second method step, with use being made of a respectively identical power and focusing but a different advancement speed.

(8) In the case of this procedure, the quality of the weld seam can be improved, but this method is relatively slow. In addition, having the same laser beam travel over the weld seam multiple times at different advancement speeds is expensive and complicated to carry out.

(9) Various methods for laser welding coated workpieces have also been disclosed in JP 2003 094 184 A, JP 2002 178 178 A, JP 2002 219 590 A, JP 2009 050 894 A, WO 2007/060479 A1, WO 03/031 111 A1, JP 2002 160 083 A and JP H04 231 190 A.

(10) DE 10 2010 003 750 A1 discloses altering the beam profile characteristic of a laser beam by means of a multicladd fiber.

SUMMARY

(11) In an embodiment, the present disclosure provides a method for laser welding two coated workpieces that includes positioning an upper workpiece and a lower workpiece on top of each other and passing a first laser beam over the upper and lower workpieces from a side of the upper workpiece so as to at least partially evaporate the respective coating of each of the workpieces on their facing sides along a depletion trace. A second laser beam is passed over the workpieces from the side of the upper workpiece so as to melt a material of the two workpieces within the depletion trace, and thereby weld the workpieces to one another. In the first laser passing, the first laser beam melts the material of the upper workpiece, so that a web of non-melted material of the upper workpiece remaining between the melted material of the upper workpiece and the facing side of the upper workpiece.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Subject matter of the present disclosure will be described in even greater detail below based on the exemplary figures. All features described and/or illustrated herein can be used alone or combined in different combinations. The features and advantages of various embodiments will become apparent by reading the following detailed description with reference to the attached drawings, which illustrate the following:

(2) FIG. 1 shows a schematic plan view of two workpieces positioned one on top of the other during an exemplary variant of the method according to the invention;

(3) FIG. 2 shows a schematic cross-sectional view of the workpieces of FIG. 1 through the sectional plane II of FIG. 1, after the first method step;

(4) FIG. 3 shows a schematic cross-sectional view of the workpieces of FIG. 1 through the sectional plane III of FIG. 1, after the second method step;

(5) FIG. 4 shows a schematic side view of an installation for carrying out the laser welding method according to the invention;

(6) FIG. 5 shows a schematic view of an apparatus for reshaping a raw laser beam, for the invention;

(7) FIG. 6 shows a schematic cross section of a reshaped laser beam, for the invention;

(8) FIG. 7 shows a schematic intensity distribution of a reshaped laser beam, along a direction y transversely with respect to the propagation direction of the reshaped laser beam;

(9) FIG. 8 shows a light-optical recording, from an experiment, of a polished section of workpieces

that were welded as per one variant of the method according to the invention, after the first method step;

(10) FIG. 9 shows a light-optical recording, from an experiment, of a polished section of the workpieces of FIG. 8 that were welded as per one variant of the method according to the invention, after the second method step; and

(11) FIGS. 10(a) and 10(b) show light-optical recordings, from an experiment, in a plan view from above (image at the top, marked 10(a)) and from below (image at the bottom, marked 10(b)) of the final weld seam of the workpieces of FIG. 9.

DETAILED DESCRIPTION

(12) The object of the invention is to provide a method for laser welding coated workpieces that are positioned one on top of the other, by means of which method a high quality of the weld seam can be achieved and which method can be carried out more quickly and easily.

(13) This object is achieved according to the invention by means of a method of the type mentioned in the introduction which is distinguished by the fact that, in the first method step, the material of the upper workpiece is melted by the laser beam, with a web of non-melted material of the upper workpiece remaining between the melted material of the upper workpiece and the side of the upper workpiece that faces the lower workpiece.

(14) In the context of the present invention, it is provided not just to heat the upper workpiece that faces the laser beam in the first method step, but also to melt it over part of its depth. This makes it possible to introduce the heat required to at least partially evaporate the coating at the joint of the overlapping workpieces positioned one on top of the other more quickly than when merely effecting heating without (even just partially) melting the upper workpiece in the course of the first method step. Correspondingly, in the first method step, higher advancement speeds are possible, and the method can be carried out more quickly overall.

(15) During the first method step, a web of non-melted material of the upper workpiece that consists of the base material of the workpiece remains, and therefore, in the first method step, the melt pool of the laser beam does not come into contact with the coating material at the joint of the two workpieces. This avoids a situation in which evaporating coating material from the joint is introduced in the depth of the melt pool or else into a vapor capillary during the first method step. Correspondingly, spatter formation caused by the coating material does not occur.

(16) In the first method step, the coating material from the joint is at least partially, and preferably virtually completely, evaporated over the width of a depletion trace by the heat of the laser beam. The evaporated coating material can in particular escape laterally outward along the joint of the workpieces. In the subsequent second method step, when the laser beam or a vapor capillary of the welding process also penetrates the lower workpiece through the depletion trace, only a little or even virtually almost no coating material from the joint enters the melt pool or the vapor capillary, as a result of which spatter or pores on account of evaporating coating material are effectively avoided. The quality of the final weld seam that can be achieved is correspondingly high.

(17) During the laser welding method according to the invention, during both the first method step and the second method step the two workpieces are positioned one on top of the other, and are in direct contact with one another in an overlapping region, in particular over their surface area. For this purpose, the two workpieces are typically clamped onto one another or pressed against one another during the first and the second method step, and typically the workpieces lie one on top of the other virtually without a gap. It should be noted that coating material that even in that case has evaporated (is gaseous) at the joint can flow along the joint, and in particular can flow out laterally. In particular, the two workpieces are not kept spaced apart one above the other during the first and the second method step.

(18) If desired, the laser beam of the first method step ("first laser beam") and/or the laser beam of the second method step ("second laser beam") may undergo beam shaping, for instance by a multiclad fiber and/or a diffractive optical element, in order to obtain a specific geometric

configuration of the laser beam. In particular for the first laser beam, a configuration with an aspect ratio ASP (ratio of the greatest width transversely with respect to the advancement direction to the greatest extent in the advancement direction) of $ASP \geq 2$, or even $ASP \geq 4$, is preferred, it being possible in particular to set approximately a linear shape, rectangular shape or an elliptical shape. In particular for the second laser beam, the application of beam shaping by means of a core focus portion and (at least) one ring focus portion (with a lower power density in the ring focus portion than in the core focus portion) has proven successful.

(19) Typically, both the first method step and the second method step are carried out as deep welding operations (with creation of a vapor capillary). In the first method step, it is in turn possible for heat to be introduced into the joint particularly quickly in this respect. In the second method step, it is possible also for heat to be introduced for melting purposes into the material of the second workpiece particularly quickly. For method steps in deep welding mode, the respective laser beam can advantageously be configured by means of a core focus portion and (at least) one ring focus portion, in order to increase the processing speed. The ring focus portion and core focus portion usually have a circular cross section; however, it is also possible to set up other annular geometries (for instance rectangular-annular geometries), in particular for the first laser beam, if a high aspect ratio is desired.

(20) The laser welding method according to the invention can be performed by passing the two laser beams of the first and second method steps over the workpieces to be welded once.

(21) The two coated workpieces are typically galvanized steel sheets. However, it is also possible to use other material combinations for the workpieces, in particular for which the evaporation temperature of the material of the coating is lower than the melting temperature of the base material of the workpieces. The workpieces typically have a thickness of 0.5-2.5 mm, usually 0.75-1.5 mm. Coatings of the workpieces typically have a thickness of 10 μm to 200 μm .

Preferred Variants of the Method According to the Invention

(22) In a preferred variant of the method according to the invention, the first method step and the second method step take place at the same time at different locations on the workpieces positioned one on top of the other, the laser beam of the first method step leading the laser beam of the second method step. This makes it possible for the method to be carried out particularly quickly.

(23) A further development of this variant in which the laser beam of the first method step and the laser beam of the second method step have the same advancement speed is advantageous. This can be carried out particularly easily; in that case, laser optical units for the first and the second laser beam can be moved together relative to the workpieces.

(24) A variant, in which the material of the first workpiece that was melted in the first method step is solidified again at least on the workpiece surface if the second method step takes place later at the same location, is preferred. This makes it possible to improve control of the method sequence and to achieve a higher quality of the weld seam. The solidified surface of the melt pool after the first method step ensures continuous calming of fluctuations in the melt pool, before the second method step begins. Correspondingly, the setup of the second method step primarily defines the melt pool dynamics during the second method step.

(25) A variant, in which the material of the first workpiece that was melted in the first method step is completely solidified again if the second method step takes place later at the same location, is likewise preferred. This likewise makes it possible to improve control over the method sequence and to achieve a higher quality of the weld seam. The completely solidified melt pool after the first method step ensures continuous calming of fluctuations in the melt pool, before the second method step begins. Correspondingly, essentially only the setup of the second method step defines the melt pool dynamics during the second method step. Furthermore, it is always the “full” enthalpy of melting that is expended to create the melt pool in the second method step or to progress it, thereby likewise improving control over the method sequence.

(26) Particularly preferred is a variant in which, in the first method step, the material of the upper

workpiece is melted to a great enough extent that, for a maximum melting depth AT of the melted material of the upper workpiece in comparison with the thickness DW of the upper workpiece, the following applies:

$60\% \leq AT/DW \leq 95\%$,

preferably $70\% \leq AT/DW \leq 90\%$.

(27) These melting depths allow heat to readily access the joint, in order to bring about evaporation of the coating at the joint or on the facing sides of the workpieces in the depletion trace. In the case of conventional workpiece thicknesses, as a safeguard, the remaining height of the web is enough to avoid coating material accessing the melt pool or a vapor capillary in the first method step.

(28) Particularly preferred is a variant in which, for a maximum width $B1$ of the laser beam in the first method step and a maximum width $B2$ of the laser beam in the second method step, in each case measured transversely with respect to the advancement direction and on the surface of the upper workpiece, the following applies:

$B1 \geq 2 * B2$,

preferably $B1 \geq 3 * B2$,

particularly preferably $B1 \geq 5 * B2$. A comparatively wide laser beam of the first method step makes it possible firstly to create a correspondingly wide depletion trace, in particular compared to the width of the later fusion zone of the workpieces. In addition, a large width of the first laser beam and a correspondingly wider distribution of the laser power radiated in reduces the risk of penetration occurring with the melt pool as far as the joint of the workpieces in the first method step, in particular in the event of inhomogeneities in the workpiece or at the joint. At the same time, the creation of a (sufficiently wide) depletion zone is highly reliable, with the result that the introduction of coating material from the joint in the second method step can similarly be reliably prevented.

(29) Also preferred is a variant which provides that, for an average laser power $P1$ of the laser beam in the first method step and an average laser power $P2$ of the laser beam in the second method step, the following applies:

$P1 \geq 1.5 * P2$,

preferably $P1 \geq 3 * P2$. The comparatively high laser power $P1$ of the first laser beam makes it possible to introduce a lot of energy into the first workpiece or into the joint in a short time, with the result that the coating material in the joint can evaporate in a short time, and correspondingly high advancement speeds in the first method step are possible. It will be noted that the (average) power density of the first laser beam is typically considerably lower than the (average) power density of the second laser beam, typically by a factor of 2 or more, usually by a factor of 4 or more.

(30) In an advantageous variant, for a focal position $F1$ of the laser beam of the first method step and a focal position $F2$ of the laser beam of the second method step, the following applies:

$F1 < F2$,

preferably $F1 < F2 \leq 0$. This has proven successful in practice in order, in the method, to bring about a favorable distribution of the laser power radiated in, and in particular in order to set the formation and depth of the vapor capillaries in the two method steps. A negative focal position indicates that the focus is below the workpiece surface of the upper workpiece (that is to say in the workpiece).

(31) Particularly preferred is a variant in which, for a width BE of the depletion trace on the facing sides of the workpieces after the first method step and a width BNK of the melted material in a joint plane of the facing sides of the two workpieces after the second method step, in each case measured transversely with respect to an advancement direction, the following applies:

$BE \geq 2 * BNK$,

preferably $BE \geq 3 * BNK$. With these relationships, the introduction of coating material into the melt pool in the second method step can be minimized and this can also be ensured with high reliability in the event of inhomogeneities, and correspondingly it is possible to achieve a very good quality

of the weld seam.

(32) Furthermore advantageous is a variant in which the laser beam of the first method step and/or the laser beam of the second method step is a reshaped laser beam and comprises a core focus portion and at least one ring focus portion, the core focus portion and the ring focus portion being coaxial with one another and the ring focus portion surrounding the core focus portion. This beam shaping has proven successful to avoid weld seam defects, both in the first method step but also and primarily in the second method step, in particular in deep welding mode. This beam shaping makes it possible to achieve calming of the melt pool dynamics, which, together with the avoidance according to the invention of the introduction of coating material from the joint into the melt pool ensures especially high quality of the weld seam and the same time allows a quick method sequence (in particular at high advancement speed in the two method steps). This variant is preferably applied for laser beams reshaped with a circular cross section, but can also be applied for non-circular, reshaped laser beams, for example rectangular, reshaped laser beams.

(33) In this respect, particularly preferred is a further development which provides that the reshaped laser beam is created by means of a multicladd fiber, a first part of the raw laser beam being fed into a core fiber of the multicladd fiber and at least one second part of the raw laser beam being fed into a ring fiber of the multicladd fiber at an input end, the ring fiber surrounding the core fiber, and the reshaped laser beam being obtained at an output end of the multicladd fiber. In this way, beam shaping can be comparatively easily carried out using a ring focus portion and a core focus portion, and in addition the division between these two portions (provided it is necessary) can be altered relatively easily or the respective welding task can be adapted by altering the division of the raw laser beam between the first part and the second part.

(34) Particularly preferred is a variant in which the laser beam of the first method step and the laser beam of the second method step are created using different laser sources, and are directed onto the workpieces positioned one on top of the other using different laser optical units, in particular with the two laser optical units being arranged at a fixed distance in relation to one another. In this variant, it is especially easily possible to independently select or set the laser power and the beam shaping of the first and the second laser beam, and in addition it is easily possible to apply the first and the second laser beam at the same time and independently. Arranging the laser optical units at a fixed distance easily makes it possible also to set up a fixed distance between the processing zones of the two laser beams on the workpieces positioned one on top of the other. In this instance, laser source means a structural unit composed of laser medium, pump and laser resonator.

(35) Furthermore advantageous is a variant in which the workpieces positioned one on top of the other are welded by full penetration welding in the second method step. This makes it possible firstly to achieve a particularly fixed connection between the workpieces, and secondly makes it possible to feed in a filler wire from below, if desired.

(36) A preferred variant of the method according to the invention provides that the workpieces are galvanized steel sheets with a thickness DW , with $0.5 \text{ mm} \leq DW \leq 1.8 \text{ mm}$, preferably $0.7 \text{ mm} \leq DW \leq 1.5 \text{ mm}$, that an average laser power $P1$ of the laser beam in the first method step is selected with $6 \text{ kW} \leq P1 \leq 20 \text{ kW}$, preferably $8 \text{ kW} \leq P1 \leq 16 \text{ kW}$, and an average laser power $P2$ of the laser beam in the second method step is selected as $3 \text{ kW} \leq P2 \leq 6 \text{ kW}$, that a maximum width $B1$ of the laser beam of the first method step and a maximum width $B2$ of the laser beam of the second method step on the surface of the upper workpiece, in each case measured transversely with respect to the advancement direction, are selected with $2.5 \text{ mm} \leq B1 \leq 5 \text{ mm}$ and $0.25 \text{ mm} \leq B2 \leq 0.75 \text{ mm}$, that an advancement speed $V1$ of the laser beam of the first method step and an advancement speed $V2$ of the laser beam of the second method step are selected with $V1 \geq 4 \text{ m/min}$ and $V2 \geq 4 \text{ m/min}$, preferably $V1 \geq 6 \text{ m/min}$ and $V2 \geq 6 \text{ m/min}$, particularly preferably with $V1 = V2$ and $6 \text{ m/min} \leq V1 \leq 20 \text{ m/min}$ and $6 \text{ m/min} \leq V2 \leq 20 \text{ m/min}$, and that a spatial distance ΔX between the laser beams of the first and the second method step is selected with $\Delta X \geq 50 \text{ mm}$ and/or an interval time TP between the action of the laser beams of the first and the second method step is selected with $TP \geq 0.50 \text{ s}$. In

practice, these parameters have led to welded connections with a high weld seam quality.

(37) The use of a method according to the invention that was described above for welding motor vehicle bodywork components also falls within the scope of the present invention. In other words, the two workpieces (in the course of or after being welded) are installed in the bodywork of a motor vehicle. In motor vehicles, coated metal sheets for which a good weld seam quality is desired, this weld seam quality being readily achievable by the present invention, are frequently used for corrosion protection purposes.

(38) Further advantages of the invention will become apparent from the description and the drawing. Likewise, according to the invention, the features mentioned above and those that will be explained still further can be used in each case individually by themselves or as a plurality in any desired combinations. The embodiments shown and described should not be understood as an exhaustive enumeration, but rather are of illustrative character for outlining the invention.

(39) FIG. 1 shows a schematic plan view of the laser welding of two coated workpieces **101**, **102** according to an exemplary variant of the method according to the invention.

(40) The workpieces **101** and **102** are positioned one on top of the other, i.e. lie directly one on top of the other, in an overlap region **103**. In the variant shown, the workpiece **101** is arranged at the top, and the workpiece **102** is arranged at the bottom (and correspondingly shown partially in dashed lines, where it is overlapped by the upper workpiece **101**). In the overlapping region **103**, the two workpieces are in mutual contact over their surface areas (touching contact); the workpieces are typically mechanically clamped (pressed) onto one another throughout the method (not shown in any more detail, but cf. FIG. 4 in this respect). The two workpieces **101**, **102** are in the form of zinc-coated steel sheets in this instance (cf. FIG. 2 and FIG. 3 in this respect), which are to be installed in a bodywork of a motor vehicle. The workpieces **101**, **102** are welded to one another along an advancement direction VR using two laser beams **104**, **105**, which are directed onto the workpiece surface **106** of the upper workpiece **101**. The laser beams **104**, **105** move in the advancement direction VR relative to the workpieces **101**, **102** along a (planned) weld seam **100** in the overlap region **103**.

(41) In a first method step, first of all the upper workpiece **101** is irradiated with the laser beam **104** (first laser beam), which is at the front with respect to the advancement direction VR, in the overlap region **103**. The first laser beam **104** has a width B1 on the workpiece surface **106** and in this instance has an approximately rectangular cross section (on the workpiece surface **106**) and has an aspect ratio ASP of approximately $ASP=6.5$ in this instance; the width B1 of the first laser beam **104** transversely with respect to the advancement direction VR is greater than its extent in the advancement direction here. Behind it, the first laser beam **104** creates a trail-like melt pool **107** of liquid, melted workpiece material. Behind the melt pool **107**, the workpiece material is completely solidified again within the boundaries of two external weld seam edges **108a**, **108b**.

(42) In a second method step, the upper workpiece **101** is then irradiated with the laser beam **105** (second laser beam), which is at the rear with respect to the advancement direction VR, in the overlap region **103**, which laser beam has a width B2 on the workpiece surface **106** and in this instance has an approximately circular cross section (on the workpiece surface **106**). In this respect, in the variant shown, it approximately holds true that $B1=3.5*B2$, the widths being measured transversely with respect to the advancement direction VR on the workpiece surface **106**. The second laser beam **105** in this instance has a core focus and a ring focus that surrounds it (for more detail in this respect, see FIGS. 5-7). Behind it, the second laser beam **105** in turn creates a trail-like melt pool **109** of liquid, melted workpiece material. Behind the melt pool **107**, the workpiece material is completely solidified again within the boundaries of two internal weld seam edges **110a**, **110b**.

(43) The weld seam **100** of the two workpieces **101**, **102** that is to be welded is passed over in each portion by the two laser beams **104**, **105** in succession, in the exemplary variant shown the two laser beams **104**, **105** (provided the end of the weld seam **100** has not been reached) being applied

at the same time and having the same advancement speed $V1=V2$, and correspondingly having a constant distance ΔX (measured from center to center of the laser beams **104**, **105** on the workpiece surface **106**) in relation to one another along the weld seam **100**. In the variant shown, this distance ΔX is large enough that the second laser beam **105** always penetrates workpiece material that has completely solidified again, i.e. the two melt pools **107**, **109** are separate from one another. For this purpose, the distance ΔX is typically greater than 50 mm. An interval time is therefore provided between the two laser processing operations of the first method step and of the second method step, which interval time allows the workpieces to cool down in the meantime; a typical interval time is 0.5 seconds or more.

(44) FIG. 2 shows a schematic cross section (perpendicularly with respect to the advancement direction) through the workpieces **101**, **102** of FIG. 1 at plane II, that is to say after the first method step.

(45) For each of the workpieces **101**, **102**, coatings **101a**, **102a** on the respective top side, coatings **101b**, **102b** on the respective bottom side, and basic bodies **101c**, **102c**, to which the coatings **101a**, **101b**, **102a**, **102b** are applied can readily be seen in the cross section. The basic bodies **101c**, **102c** are manufactured from steel, and the coatings **101a**, **101b**, **102a**, **102b** from zinc. In the variant shown, the workpieces **101**, **102** have similar forms, in particular with the same thickness DW.

(46) The first laser beam **104** (depicted as a dashed line, because it lies behind the plane of the drawing) has a negative focal position F1 with respect to the workpiece surface **106**, with its focus **104a** in the bottom half of the lower workpiece **102** in this instance. The first laser beam **104** has melted the workpiece material in the region of the melt pool **107** (which in this instance has solidified again). The melt pool **107** extends only in the upper workpiece **101** and has a (maximum) melting depth AT which in this instance is approximately 75% of the thickness DW of the upper workpiece **101**. A web **114** of material of the first workpiece **101** that was not melted by the first laser beam remains between the melt pool **107** and the bottom side of the upper workpiece **101**.

(47) The heating of the upper workpiece **101** by the first laser beam **104** has also resulted in a depletion trace **112** (marked by hatching) being produced at the joint **111** of the workpieces **101**, **102**, which is the contact region of the facing sides of the workpieces **101**, **102** positioned one on top of the other and resting on one another, in which depletion trace the material of the coatings **101b** and **102a** has at least partially, preferably completely, evaporated and escaped from the joint **111**. The depletion trace **112** has a width BE, it holding true in this instance approximately that $BE=0.9*B1$. In general, it preferably holds true that $BE \geq 0.5*B1$.

(48) FIG. 3 shows a schematic cross section (perpendicularly with respect to the advancement direction) through the workpieces **101**, **102** of FIG. 1 at plane III, that is to say after the second method step.

(49) The second laser beam **105** (depicted as a dashed line, because it lies behind the plane of the drawing) likewise has a negative focal position F2 with respect to the workpiece surface **106**, with its focus **105a** in the top half of the lower workpiece **102** in this instance. The focal position F2 of the second laser beam **105** thus lies somewhat further upward (closer to the workpiece surface **106**) than the focal position F1 of the first laser beam **104**. The second laser beam **105** has melted the workpiece material in the region of the melt pool **109** (which in this instance has solidified again). The melt pool **109** extends completely through the upper workpiece **101** and in this instance also completely through the lower workpiece **102**, corresponding to full penetration welding.

(50) For a width BNK of the melted material of the (solidified) melt pool **109** in the joint plane **113** of the joint **111**, it holds true here approximately that $BNK=0.33*BE$. In general, it preferably holds true that $BNK \leq 0.5*BE$. The melt pool **109** lies in the region of the joint **111**, that is to say within the depletion trace **112**, with the result that no material or only very little material of the coatings **101b**, **102a** enters the melt pool **109** or even a vapor capillary (not illustrated) created in the second method step using the second laser beam **105**. This avoids defects at the weld seam caused by abruptly evaporating material of the coatings **101b**, **102a**.

(51) FIG. 4 shows a schematic side view of an exemplary installation **400** for carrying out a welding method according to the invention.

(52) Two workpieces **101**, **102**, which in this instance are sheet-like, are positioned one on top of the other and are pressed onto one another using a clamping system **401**, in this instance comprising movable press punches **402** and supports **403**, in order to obtain approximately gap-free mutual contact over their surface area. The workpieces **101**, **102** are welded to one another along an advancement direction VR using a first laser beam **104** and a second laser beam **105** (cf. also FIGS. 1-3 in this respect).

(53) The first laser beam **104** for the first method step is created by a first laser source **404**, and the second laser beam **105** for the second method step is created by a separate second laser source **405**; the laser sources **404**, **405** each have a laser resonator, laser medium and pump (not illustrated in more detail). In the variant shown, the first laser source has a power P1 and the second laser source has a power P2, in this instance with $P1=3 \cdot P2$; however, other variants are also possible, for example also with $P1=P2$. The laser light generated in the respective laser source **404**, **405** is in each case conducted by way of a flexible light guide **406**, **407** to a respective laser optical unit **408**, **409**, which then directs the respective laser beam **104**, **105** onto the (upper) workpiece surface **106** of the upper workpiece **101**. Depending on the usage situation, respective beam shaping may be provided for the laser beams **104**, **105**, for instance set up by the light guides **406**, **407** (also see FIGS. 5-7 in this respect) and/or the laser optical units **408**, **409**, which may contain diffractive optical elements (not illustrated in more detail), for example.

(54) The two laser optical units **408**, **409** in this instance are fixedly arranged on a common mount **410**, which can be moved parallel to the advancement direction VR on a guide **412** by means of a motorized drive **411**. Correspondingly, the two laser optical units **408**, **409** are at a fixed distance in relation to one another with respect to the advancement direction VR. In this instance, the guide **412** is formed with a toothset (not illustrated in more detail) for the engagement of a toothed wheel **411a** of the motorized drive **411**. By moving the common mount **410** on the guide **412**, the laser beams **104**, **105** can be moved together with respect to the (in this instance stationary) workpieces **101**, **102**, in order to set up the advancement of the welding processes.

(55) FIG. 5 schematically illustrates an exemplary apparatus **1**, which can be used for beam shaping within the scope of the present invention for the first and/or the second laser beam (see FIGS. 1-4).

(56) A laser source **3** generates a raw laser beam **2**, which in this instance is directed via a focusing lens **5** onto the input end **4a** of a multiclاد fiber **4**, here a 2 in 1 fiber; cf. the propagation direction AR. The multiclاد fiber **4** serves as a light guide.

(57) A first part **2a** of the cross section of the raw laser beam **2** is fed into a core fiber **8** of the multiclاد fiber **4** without deflection. By means of a wedge plate **6**, a second part **2b** of the cross section of the raw laser beam **2** is deflected and fed into a ring fiber **7** of the multiclاد fiber **4**.

(58) In the multiclاد fiber **4**, the laser power of the two parts **2a**, **2b** is distributed uniformly in each case in the course of advancing in the ring fiber **7** and in the core fiber **8**; it should be noted that the length of the multiclاد fiber **4** is illustrated in a shortened manner in FIG. 5. At an output end **4b** of the multiclاد fiber **4**, a reshaped laser beam is provided (not illustrated in more detail, but cf. FIG. 6 and FIG. 7 in this respect); the reshaped laser beam is typically imaged onto the workpieces to be welded via a lens system, usually comprising a collimation lens and a focusing lens (likewise not illustrated in more detail).

(59) For setting the parts **2a**, **2b** or the power proportions for the core focus portion and the ring focus portion, the wedge plate **6** here can be moved in the direction R transversely with respect to the propagation direction AR.

(60) FIG. 6 illustrates the beam cross section of the reshaped laser beam **10** such as can be observed for example at the workpiece surface of the irradiated upper workpiece. The reshaped laser beam **10** has a core focus portion **11** surrounded coaxially by a ring focus portion **12**. A region

13 with lower laser intensity typically lies between the core focus portion **11** and the ring focus portion **12**. In this instance, the diameter DR of the ring focus portion **12** is approximately 3.5 times the diameter DK of the core focus portion **11**; it preferably holds true that $2 \leq DR/DK \leq 5$.

(61) FIG. 7 illustrates a typical intensity distribution of the reshaped laser beam **10** in cross section along a direction y leading through the center of the cross section of the reshaped laser beam **10**; the local laser intensity is plotted upward, and the location along the direction y toward the right.

(62) Within the core focus portion **11** the laser power is approximately constant, typically with a fluctuation of a maximum of 25% around the average value in the core focus portion **11**. Likewise, the laser power in the region of the ring focus portion **12** is approximately constant, typically likewise with a fluctuation of a maximum of 25% around the average value in the ring focus portion. It is readily discernible in the diagram that the (average) intensity of the reshaped laser beam **10** in the region of the core focus portion **11** is significantly greater than that in the region of the ring focus portion **12**, here approximately 8 times greater. With the geometry present here, approximately 50% of the total laser power is allotted to the core focus portion **11**, and the remainder, here likewise approximately 50% of the total laser power, is allotted to the ring focus portion. Preferably, between 45% and 85% of the total laser power of the laser beam is allotted to the core focus portion.

(63) It should be noted that a plurality of concentric ring focus portions can also be provided in the beam cross section of the reshaped laser beam, in particular by a multicladd fiber with two or more ring fibers being used for the reshaping (not illustrated in more detail).

(64) FIG. 8 and FIG. 9 each show a light-optical recording, from an experiment, of a polished section of two workpieces that were welded according to the invention, in a state after the first method step in FIG. 8 and after the second method step in FIG. 9. The workpieces each had a thickness of approximately 1 mm and were manufactured from sheet steel as basic body and provided with a zinc coating on both sides. In the first method step, a rectangular laser beam with a width B1 of 2.95 mm was applied, the laser power amounted to 6 kW, and the advancement speed 4.2 m/min. In the second method step, a circular laser beam reshaped by a 2 in 1 fiber and having a width B2 of 0.44 mm (ring focus 0.44 mm, core focus 0.11 mm, power proportion of the core focus 70%) was applied, the laser power amounted in turn to 6 kW, and the advancement speed 12 m/min. FIG. 10 shows light-optical recordings in a plan view of the finished weld seam from above (image at the top, marked a) and from below (image at the bottom, marked b).

(65) In the first method step, the upper workpiece was melted over approximately 80% of its thickness, but the joint was not reached, in keeping with the invention. Full penetration welding was carried out in the second method step. The finished weld seam is very uniform both on the top side and on the bottom side, and despite the comparatively high advancement speed in the first method step there was virtually no welding spatter, and no porosity can be seen in the solidified melt pool of the finished weld seam.

(66) While subject matter of the present disclosure has been illustrated and described in detail in the drawings and foregoing description, such illustration and description are to be considered illustrative or exemplary and not restrictive. Any statement made herein characterizing the invention is also to be considered illustrative or exemplary and not restrictive as the invention is defined by the claims. It will be understood that changes and modifications may be made, by those of ordinary skill in the art, within the scope of the following claims, which may include any combination of features from different embodiments described above.

(67) The terms used in the claims should be construed to have the broadest reasonable interpretation consistent with the foregoing description. For example, the use of the article “a” or “the” in introducing an element should not be interpreted as being exclusive of a plurality of elements. Likewise, the recitation of “or” should be interpreted as being inclusive, such that the recitation of “A or B” is not exclusive of “A and B,” unless it is clear from the context or the foregoing description that only one of A and B is intended. Further, the recitation of “at least one of

A, B and C” should be interpreted as one or more of a group of elements consisting of A, B and C, and should not be interpreted as requiring at least one of each of the listed elements A, B and C, regardless of whether A, B and C are related as categories or otherwise. Moreover, the recitation of “A, B and/or C” or “at least one of A, B or C” should be interpreted as including any singular entity from the listed elements, e.g., A, any subset from the listed elements, e.g., A and B, or the entire list of elements A, B and C.

LIST OF REFERENCE SIGNS

(68) **1** Apparatus **2** Raw laser beam **2a** First part (raw laser beam) **2b** Second part (raw laser beam) **3** Laser source **4** Multiclad fiber **4a** Input end **4b** Output end **5** Focusing lens **6** Wedge plate **7** Ring fiber **8** Core fiber **10** Reshaped laser beam **11** Core focus portion **12** Ring focus portion **13** Region with lower laser intensity **100** Weld seam **101** Upper workpiece **101a** Coating (top side of upper workpiece) **101b** Coating (bottom side of upper workpiece) **101c** Basic body (upper workpiece) **102** Lower workpiece **102a** Coating (top side of lower workpiece) **102b** Coating (bottom side of lower workpiece) **102c** Basic body (lower workpiece) **103** Overlap region **104** (First) laser beam **104a** Focus (first laser beam) **105** (Second) laser beam **105a** Focus (second laser beam) **106** Workpiece surface **107** Melt pool (created by first laser beam) **108a**, **108b** Outer weld seam edges **109** Melt pool (created by second laser beam) **110a**, **110b** Inner weld seam edges **111** Joint **112** Depletion trace **113** Joint plane **114** Web **400** Installation **401** Clamping system **402** Press punch **403** Support **404** Laser source (first laser beam) **405** Laser source (second laser beam) **406**, **407** Light guides **408**, **409** Laser optical unit **410** Common mount **411** Motorized drive **411a** Toothed wheel **412** Guide AR Propagation direction AT Melting depth BE Width of depletion trace BNK Width of the melted material in the joint plane B1 Width of first laser beam B2 Width of second laser beam F1 Focal position of first laser beam F2 Focal position of second laser beam P1 Laser power (first laser beam) P2 Laser power (second laser beam) R Direction (transversely with respect to the propagation direction of the raw laser beam) VR Feed direction V1 Advancement direction (first laser beam) V2 Advancement direction (second laser beam) y Direction (transversely with respect to the propagation direction of the reshaped laser beam) ΔX Distance between the laser beams

Claims

1. A method for laser welding two coated workpieces, comprising: positioning an upper workpiece and a lower workpiece one on top of the other, in a first laser passing, passing a first laser beam over the upper and lower workpieces from a side of the upper workpiece so as to at least partially evaporate the respective coating of each of the workpieces on their facing sides along a depletion trace, and in a second laser passing, passing a second laser beam over the workpieces from the side of the upper workpiece so as to melt a material of the two workpieces within the depletion trace, and thereby weld the workpieces to one another, wherein, in the first laser passing, the first laser beam melts the material of the upper workpiece and does not melt through the upper workpiece, so that a web of non-melted material of the upper workpiece remaining between the melted material of the upper workpiece and the facing side of the upper workpiece.
2. The method as claimed in claim 1, wherein the first laser passing and the second laser passing take place at the same time at different locations on the workpieces, the first laser beam leading the second laser beam.
3. The method as claimed in claim 2, wherein the first laser beam and the second laser beam have a same advancement speed.
4. The method as claimed in claim 1, further comprising solidifying at least on a surface of the upper workpiece the material of the upper workpiece that was melted in the first laser passing when the second laser passing takes place later at the same location.
5. The method as claimed in claim 1, further comprising completely solidifying the material of the

upper workpiece that was melted in the first laser passing when the second laser passing takes place later at the same location.

6. The method as claimed in claim 1, wherein, in the first laser passing, the material of the upper workpiece is melted to a great enough extent that, for a maximum melting depth AT of the melted material of the upper workpiece in comparison with the thickness DW of the upper workpiece, the following applies:

$$60\% \leq AT/DW \leq 95\%.$$

7. The method as claimed in claim 1, wherein, for a maximum width $B1$ of the first laser beam and a maximum width $B2$ of the second laser beam, in each case measured transversely with respect to an advancement direction and on the surface of the upper workpiece, the following applies:

$$B1 \geq 2 * B2.$$

8. The method as claimed in claim 1, wherein, for an average laser power $P1$ of the first laser beam and an average laser power $P2$ of the second laser beam, the following applies:

$$P1 \geq 1.5 * P2.$$

9. The method as claimed in claim 1, wherein, for a focal position $F1$ of the first laser beam and a focal position $F2$ of the second laser beam, the following applies:

$$F1 < F2.$$

10. The method as claimed in claim 1, wherein, for a width BE of the depletion trace on the facing sides of the workpieces after the first laser passing and a width BNK of the melted material in a joint plane of the facing sides of the two workpieces after the second laser passing, in each case measured transversely with respect to an advancement direction, the following applies:

$$BE \geq 2 * BNK.$$

11. The method as claimed in claim 1, wherein the first laser beam and/or the second laser beam is a reshaped laser beam and comprises a core focus portion and at least one ring focus portion, the core focus portion and the ring focus portion being coaxial with one another and the ring focus portion surrounding the core focus portion.

12. The method as claimed in claim 11, wherein the reshaped laser beam is passed through a multiclاد fiber, a first part of a raw laser beam being fed into a core fiber of the multiclاد fiber and at least one second part of the raw laser beam being fed into a ring fiber of the multiclاد fiber at an input end, the ring fiber surrounding the core fiber, and the reshaped laser beam being obtained at an output end of the multiclاد fiber.

13. The method as claimed in claim 1, wherein the first laser beam and the second laser beam are created using different laser sources, and are directed onto the workpieces using different laser optical units.

14. The method as claimed in claim 1, wherein the workpieces are welded by full penetration welding in the second laser passing.

15. The method as claimed in claim 1, wherein the workpieces include galvanized steel sheets with a thickness DW , with $0.5 \text{ mm} \leq DW \leq 1.8 \text{ mm}$, that an average laser power $P1$ of the first laser beam is selected with $6 \text{ kW} \leq P1 \leq 20 \text{ kW}$, and an average laser power $P2$ of the second laser beam is selected as $3 \text{ kW} \leq P2 \leq 6 \text{ kW}$, a maximum width $B1$ of the first laser beam and a maximum width $B2$ of the second laser beam on the surface of the upper workpiece, in each case measured transversely with respect to the advancement direction, are selected with $2.5 \text{ mm} \leq B1 \leq 5 \text{ mm}$ and $0.25 \text{ mm} \leq B2 \leq 0.75 \text{ mm}$, an advancement speed $V1$ of the first laser beam and an advancement speed $V2$ of the second laser beam are selected with $V1 \geq 4 \text{ m/min}$ and $V2 \geq 4 \text{ m/min}$, and a spatial distance ΔX between the first and second laser beams is selected with $\Delta X \geq 50 \text{ mm}$ and/or an interval time TP between the action of the first and second laser beams is selected with $TP \geq 0.50 \text{ s}$.

16. The method as claimed in claim 1, wherein upper and the lower workpieces comprise steel sheets coated with zinc.

17. The method as claimed in claim 1, wherein the upper and lower workpieces are motor vehicle bodywork components.

